Unbounded Knapsack Problem

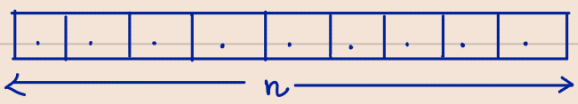
Solution Description : The basic approach to the problem remains the same as the 0/1 Knapsack Problem, with a slight modification that after taking the item i in the Knapsack, unlike 0/1 Knapsack algorithm, we will not increment the index i. As a result, we will always have the scope of taking item i in the Knapsack more than once.

Recurrance Relation :

$$\text{Knapsack}(i, w) = \begin{cases} \text{Knapsack}(i+1, w) & \text{if } wt[i] > w \\ \max \begin{Bmatrix} \text{Knapsack}(i+1, w) \\ Ct[i] + \text{Knapsack}(i, w - wt[i]) \end{Bmatrix} & \text{otherwise} \end{cases}$$

Note : Unlike 0/1 Knapsack, we do not need a 2D Array for the DP version of the Solution.

Dynamic Programming :

$M =$ 

$n =$ capacity

$$M[w] = \text{for } i \in [1, n] \left\{ \max\left(M[w], M[w - wt[i]] + Pf[i] \right) \right\}$$
$$\forall\ w \in [1, n]$$

2D Realization of Unbounded Knapsack :

The 2D realization of the problem can be given with the following relation :

$$uks(i, w') = \begin{cases} uks(i-1, w') & \text{if } w[i] > w' \\ \max\{ uks(i-1, w'),\ P[i] + uks(i, w' - w[i]) & \text{if } w[i] \le w' \end{cases}$$

Number of Nodes in a Balanced Binary Tree of height 'h'

Number of nodes in a balanced binary tree of height 1 = 1

Number of nodes in a balanced binary tree of height 2 = 2

Number of nodes in a balanced binary tree of height 3 = 4

In order to find the number of nodes in a binary tree of height 4, we will take the balanced binary tree of height 3 and add a new root-node to it.

Now, this newly created tree is unbalanced at the root node. Thus to balance the root node, we need to argument the root node with another with another balance tree with height of 2.

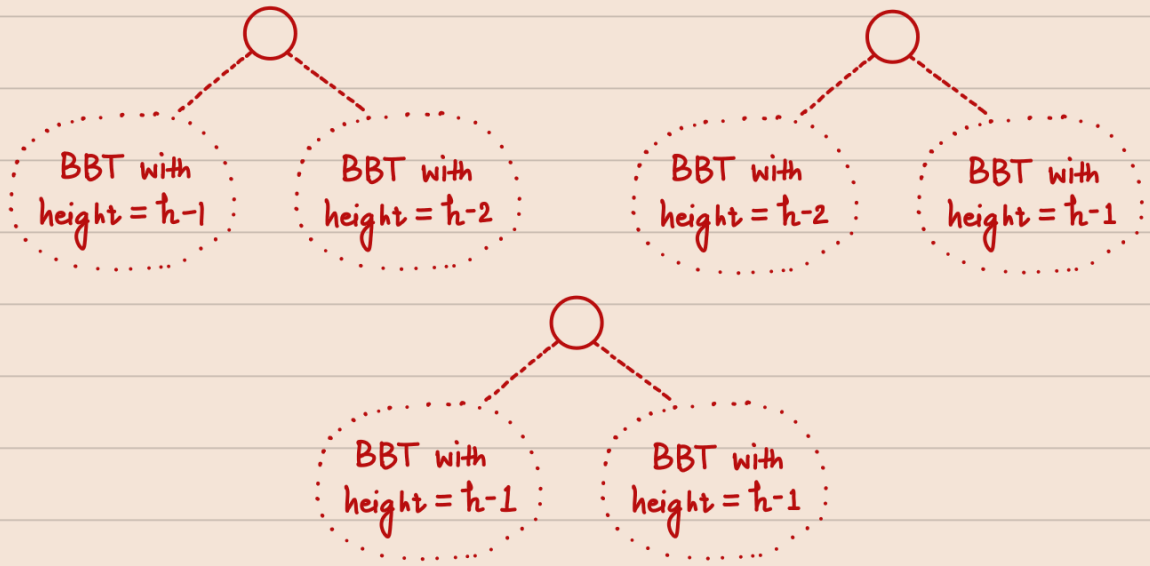
Similarly, for constructing a height balanced binary tree of height 5, we need to add a root node to the height balanced binary tree of height 4 and balance the root node with a height balanced binary tree of height 3.

If $M[h]$ is the no. of nodes in a height balanced tree of height h then,

$$M[h] = M[h-1] + 1 + M[h-2]$$

Number of Balanced Binary Tree of height 'h'

Number of possible balanced binary trees of height h can be one of the following three forms,



Let N[h] represent the number of BBT of height h, then,

$$N[h] = N[h-1] \cdot N[h-2] + N[h-2] \cdot N[h-1] + N[h-1] \cdot N[h-1]$$

or, $$N[h] = 2\,N[h-1] \cdot N[h-2] + N[h-1]^2$$